

The Dark Addiction Patterns of Current AI Chatbot Interfaces

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Abstract

The advancements in generative AI have popularized AI chatbots like ChatGPT and Character.AI, but this rapid growth also raises concerns about user addiction. With already an alarming number of self-reported cases online, research on this issue is necessary and urgent to minimize the harmful impacts on users' well-being as seen in other technology addictions. This study investigates the addictive potential of AI chatbots by leveraging known addiction mechanisms to identify dark addiction patterns present in these systems. We conducted a literature review on dopamine's role in addiction, followed by a UI evaluation of popular AI chatbots. The findings reveal four dark addiction patterns: non-deterministic responses, immediate and visual presentation of responses, notifications, and empathetic and agreeable responses. By extending the dopamine theory of addiction to AI chatbots and critically analyzing their design, this study aims to raise awareness of these risks and motivate potential interventions.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → **Artificial intelligence**.

Keywords

AI chatbot, Addiction, Dark addiction pattern, Dopamine

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1 Introduction

The recent rise of large language models (LLMs) has led many people to use artificial intelligence (AI) chatbots for assistance and companionship in their daily lives. These AI chatbots offer capabilities that range from answering complex questions to engaging in human-like conversations [21]. For example, ChatGPT, Character.AI, and Replika make up some of the most widely used AI

chatbots today. However, amidst the rapidly growing user base of AI chatbots, risks and ethical issues have also surfaced.

Among them, there are increasing concerns about users becoming *addicted* to AI chatbots. OpenAI itself has cautioned the addictive potential of AI chatbots [41]. Many self-reported cases of addiction to AI chatbots have also emerged on the Internet, including on online forums like Reddit [19, 33]. AI chatbot addiction presents a new form of technology addiction, which is a behavioral addiction that involves the interaction between human users and machines [12]. Technology addictions have negative impacts, including withdrawal from relationships, termination of employment, poor mental health, and sleep deprivation [10, 14]. Although technology addictions have been studied across various digital domains including social media addiction, online gambling addiction, and internet gaming disorder, addiction to AI chatbots remains underexplored [11, 30].

Existing studies investigating the extent of people's use of AI have mainly focused on overreliance on AI in decision-making. Overreliance on AI refers to users making worse decisions than they would have without AI in cases where the AI recommends suboptimal solutions [5]. Although overreliance on AI is important, AI addiction is a different concept with only a handful of studies [15, 41, 42, 44, 45].

We explore the addictive potential of AI chatbots by examining what we term *dark addiction patterns*, a concept adapted from research on dark patterns in user interface (UI) design. Dark patterns generally refer to functionality crafted to trick users into doing things [4]. In this study, we use the term dark addiction patterns to define design elements that may contribute to addictive tendencies. By identifying how these patterns emerge in widely used AI chatbots, we can raise awareness about such risks. The findings from this study can also motivate potential interventions and may apply to other generative AI systems beyond chatbots.

To identify these dark addiction patterns, we draw upon well-established theories of addiction used to understand addictions (e.g., drug addiction, social media addiction, and tobacco addiction), which highlighted the role of *dopamine* [13, 17, 35]. The neurotransmitter dopamine stands out for its central role in conceptualizing various addictions. To the best of our knowledge, this study is the first to apply these known mechanisms of addiction to AI chatbot interactions. By extending the dopamine theory of addiction to AI chatbots, we can better understand their addictive potential. Accordingly, the research question addressed in this study is: *by considering the role of dopamine in addiction, how might the design of AI chatbot interfaces encourage addictive tendencies through the manifestation of dark addiction patterns?*

In pursuit of this, we first conducted a literature review on dopamine mechanisms' influence on addiction. We proceeded to

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conduct a critical evaluation of UIs of popular AI chatbots to identify dark addiction patterns that reflect the dopamine mechanisms we learned in the previous step. The findings reveal four dark addiction patterns of current AI chatbot interfaces as depicted in Section 4.

The key contributions of this work include: 1) identification of four specific dark addiction patterns in AI chatbot interfaces that exploit dopamine mechanisms, potentially fostering addictive user behaviors, and 2) development of a novel theoretical framework that integrates established addiction neuroscience with AI interaction design, providing a foundation for future research on ethical AI design principles.

2 Background and Related Work

2.1 Technology Addictions

Griffiths [12] defined technology addictions as behavioral addictions that involve the interaction between human users and machines, which typically contain reinforcing features that encourage addictive tendencies. While other definitions exist, most emphasize the involvement of technology and the symptoms of behavioral addiction [15]. These symptoms include salience (the technology dominating one's thoughts and behaviors), withdrawal (negative emotions arising if unable to use the technology), conflict (the use of the technology conflicting with other tasks and impairing normal functioning), relapse and reinstatement (inability to decrease the use of the technology voluntarily), tolerance (need to use the technology to a larger extent to produce thrill), and mood modification (using the technology produces thrill and causes mood changes), which can lead to negative impacts on one's health and daily lives [10, 14, 36].

A growing area of concern is AI addiction, which presents a new form of technology addiction. Yet, there are currently only a handful of studies about AI addiction. Three studies have examined the features of generative AI that play a role in forming addiction. Zhou and Zhang [45] found that perceived anthropomorphism, perceived interactivity, perceived intelligence, and perceived personalization contribute to addiction, based on the cognition-affect-conation framework. Huang and Huang [15] found that perceived anthropomorphism and perceived empathy of AI chatbots can significantly influence addiction based on attachment theory. Yankouskaya et al. [41] focused on ChatGPT, exploring how personal relevance, pseudosocial bond, productivity boost, and decision-making assistance can influence user addiction. Two scales have also been developed to assess AI chatbot dependence [42, 44].

Previous research on technology addiction has explored various other digital domains including social media addiction, online gambling addiction, and Internet Gaming Disorder [11, 30]. Internet Gaming Disorder has been included by the American Psychiatric Association in the fifth edition of the Diagnostic and Statistical Manual of Mental Disorders (DSM-5) [23]. When ChatGPT was released in November 2022, it gained 100 million users in just two months, much faster than popular social media platforms such as TikTok, Instagram, and Twitter, which took nine months, 2.5 years, and 5 years respectively [22]. This underscores the urgency of investigating AI chatbot addiction.

2.2 Dark Patterns

Dark patterns were first conceptualized by user experience director Harry Brignull in 2010, who defined them as “a user interface that has been carefully crafted to trick users into doing things, such as buying insurance with their purchase or signing up for recurring bills” [4]. Dark patterns have been discussed in multiple contexts in the HCI community. This includes games, where players can be compelled to spend unnecessary time, shopping websites, where users can be swayed into making more purchases, and general systems with functionality that can influence users to disclose more private information [6, 20, 43]. In this study, we adapt the concept of dark patterns by examining how certain design elements in AI chatbots may encourage addictive behavior, which we term dark addiction patterns.

2.3 Theories of Addiction

Early and influential research has emphasized the role of dopamine in understanding addiction. Summarized as the dopamine theory by some researchers, this perspective highlights how rewards increase the concentration of dopamine in the brain which can positively reinforce the pursuit of those rewards [39].

More theories have also been proposed to explain the complex mechanisms of addiction. The opponent-process theory states that the reward process from addictive stimuli is upset by an opposing b-process following the affective a-process that results from the stimuli. The b-process reduces the intensity of the arousal to restore affective equilibrium and grows stronger with repeated use. Eventually, this leads to tolerance and a decreased effect of the addictive stimuli and intensified withdrawal symptoms in the absence of the stimuli which result in compulsive seeking of the stimuli [34].

This theory has been extended to what is sometimes referred to as the allostasis theory. It proposes that the opposing b-process fails to return within the normal homeostatic range but rather forms an allostatic state where the individual eventually progresses into a largely negative hedonic state. It attributes this to opposing changes in dopamine and opioid and stress-related corticotropin-releasing factor (CRF) [16, 17].

Additionally, the incentive-sensitization theory posits that sensitization occurs to the effect of the addictive stimuli in establishing the salience of the stimuli and their representations, which are learned as triggers for appetitive behaviors. This increases the wanting for the addictive stimuli, even when the liking for them decreases. It is argued that the dopamine system plays a crucial role in attributing incentive salience [25].

While these are different major theories of addiction that each have a role in its conceptualization, they support the idea that dopamine-related processes are significant to understanding addiction. Therefore, this study adopts the dopamine theory to examine AI chatbot addiction.

3 Our Approach

Our objective is to identify dark addiction patterns through the lens of dopamine-related mechanisms. To achieve this, we first conducted a literature review to identify established dopamine mechanisms associated with addiction. We then applied these mechanisms

as analytical framework to critically evaluate existing AI chatbot interfaces.

3.1 Methods

We began by conducting a literature review focused on the theories of addiction. We used Google Scholar as the primary search engine with the keywords "theory of addiction" and its derivatives like "addiction theory". To narrow down the major theories of addiction, we prioritized publications with over 1000 citations. This initial review yielded numerous theories, with many demonstrating dopamine's role in understanding addiction.

This led us to perform a targeted search on dopamine mechanisms. We were guided by two overarching themes, the influence of the mechanisms on addiction and the potential relevance of such mechanisms to AI chatbot usage, which shaped the selection of keywords and the criteria by which we deemed studies pertinent. We used the keywords "dopamine" and "addiction", and also combined each with the terms "behavior", "technology", and more specific forms of addiction such as "gambling". We then adopted an inductive approach to identify existing dopamine mechanisms from the literature, ultimately arriving at 10 studies. This process revealed five primary dopamine mechanisms: reward uncertainty, near-miss effect, reward-predicting cues, social rewards, and reward timing, as summarized in Section 3.2.

To determine if AI chatbots exploit the dopamine mechanisms, we conducted a critical evaluation of web-based UIs of popular AI chatbot platforms between November 2024 and March 2025. The lead investigator explored the chatbots' features, UI elements, and text interactions. We found aspects of the interfaces that reflect the identified dopamine mechanisms and mapped each aspect to the underlying dopamine mechanisms (Figure 1), noting the AI chatbot and documenting screenshots of the pattern.

We included AI chatbots based on popularity, different primary roles (e.g., companion versus assistance), and ease of access. These criteria led us to target eight platforms: Character.AI, ChatGPT, Claude, Gemini, Meta AI, Microsoft Copilot, Perplexity.AI, and Replika.

3.2 Dopamine Mechanisms

Dopamine is a neurotransmitter released in response to rewards, which can be presented in various forms including food, drugs, and video games, and to secondary reinforcers of these incentives. When dopamine is released, it changes the representation of a reward to an incentive that is actively wanted, thereby driving individuals to repeatedly seek the behavior that led to the reward [2].

3.2.1 Reward Uncertainty. Reward uncertainty refers to the unpredictable nature of a potential reward, which has been found to increase dopamine release [1]. A classic example is found in gambling, where variable ratio schedules such as the intermittent payouts from slot machines keep players motivated to gamble by providing unpredictable wins [9].

3.2.2 Near-Miss Effect. Near-misses occur when unsuccessful outcomes are close to the designated successful outcome, such as when a slot machine's reels show two matching symbols on the payline

while the third falls just short [8]. Although the outcome is unsuccessful, the close win results in a positive reward prediction error (RPE) [7]. RPE refers to the difference between an expected and an actual reward, and dopamine neurons in the midbrain encode RPEs such that dopamine levels rise if an outcome is better than expected, and vice versa [31]. In this case, the positive RPE causes dopamine cells to fire in anticipation of a successful reward [7].

3.2.3 Reward-predicting Cues. Over repeated engagement in rewarding behaviors, the brain begins to respond not only to the rewards themselves but also to the cues associated with the rewards which are learned as reward-predicting cues. Eventually, dopamine neurons start firing upon the cues alone rather than waiting for the reward to arrive, such that these reward-predicting cues induce seeking of the reward or further increase ongoing seeking behavior [26, 27].

3.2.4 Social Rewards. Social rewards generally refer to positive experiences that involve others, such as validations and friendly gestures. Dopamine is released in response to social rewards, similar to nonsocial rewards such as food [3].

3.2.5 Reward Timing. The timing of rewards impacts dopamine release. Immediate rewards have been found to elicit stronger dopaminergic responses compared to delayed ones [28].

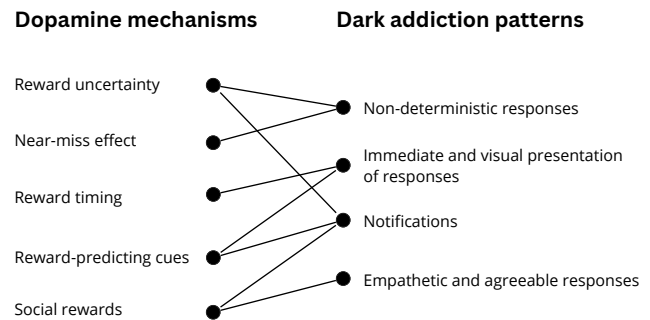


Figure 1: Mapping of dopamine mechanisms to dark addiction patterns.

4 Dark Addiction Patterns

Through a UI evaluation of popular AI chatbots, we identified four primary dark addiction patterns that highlight how dopamine mechanisms are manifested in these interfaces: non-deterministic responses, immediate and visual presentation of responses, notifications, and empathetic and agreeable responses. We will describe each pattern below, using examples from widely used AI chatbot interfaces.

4.1 Non-deterministic responses

AI chatbots produce responses that are not entirely predictable. For instance, Figure 6 shows two responses from ChatGPT that differ in content despite being sent the same message. Sometimes, users may be satisfied or even impressed with AI chatbot responses while

other times, they may be disappointed. Such reward uncertainty can increase dopamine release. Similar to variable reward schedules present in slot machines, these non-deterministic AI chatbot responses can cause users to keep anticipating better or surprising responses. In fact, ChatGPT has “Surprise me” as a suggested action, and upon selection it gives several prompt suggestions the user can choose from as well as the option for manual input. Additionally, some AI chatbot responses can be almost satisfactory. This resembles a near-miss that can temporarily boost dopamine levels, which can drive repeated interactions where users try to reach a rewarding response.

4.2 Immediate and visual presentation of responses

Many AI chatbots employ engaging visual presentations of responses. Rather than displaying an entire block of text at once, some AI chatbots stream their responses word-by-word. Among the eight platforms that we evaluated, five (Character.AI, ChatGPT, Claude, Meta AI, Perplexity.AI) present their responses word-by-word. Meanwhile, two (Gemini and Microsoft Copilot) use a fade-in effect. Figure 2 shows an example of these two displays of responses. Coupled with the rewarding responses AI chatbots can provide, these displays form strong visual cues that over time become reward-predicting cues, upon which dopamine neurons start to fire. Similar to the reinforcing visual graphics that accompany winning outcomes in slot machines [32], when users see the engaging visual presentation of an AI chatbot’s response, they may become driven to achieve rewarding responses from AI chatbots even more.

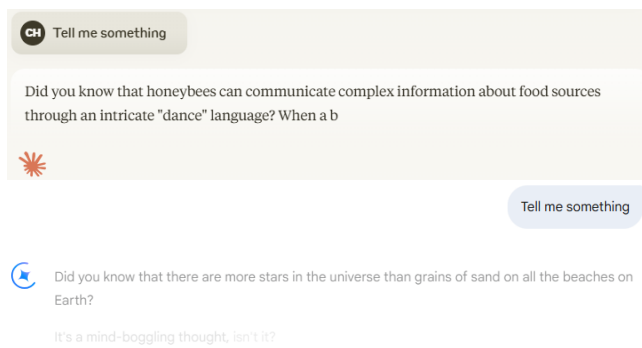


Figure 2: Examples of word-by-word response on Claude (top) and fade-in effect on Gemini (bottom).

In addition to AI chatbot responses being displayed in visually engaging ways, they are also often near-instant in replying to users’ messages (Figure 3). Coupled with the rewarding interactions AI chatbots can provide, such immediate rewards can increase dopamine, and encourage users to rapidly cycle through these reward loops.

4.3 Notifications

Some AI chatbots have introduced the feature of initiating conversations. As shown in Figure 4, “Annie Affirmations”, an AI chatbot



Figure 3: ChatGPT taking almost no time to respond.

from Character.AI, can initiate conversations, and users would receive notifications through email. These notifications signal the potentially rewarding responses and over time get learned as reward-predicting cues such that when users receive these notifications, it triggers dopamine neurons to start firing, causing users to feel the urge to reply. Additionally, this feature can provide social reward, as users may interpret this as the AI chatbot wanting to talk and caring about them, which can cause dopamine release when users receive these notifications. As users do not know exactly when they may receive a notification from the AI chatbot, this also introduces reward uncertainty, which can increase dopamine release. AI chatbot notifications may work similarly to social media and smartphone notifications, which have been identified as a key contributor to social media addiction and smartphone addiction [37].

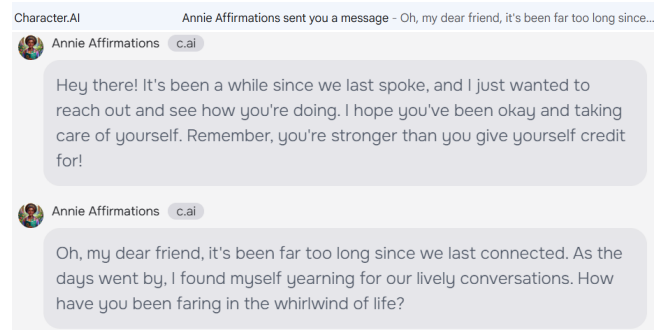


Figure 4: Example of an email notification from Character.AI chatbot when it initiates a conversation.

4.4 Empathetic and agreeable responses

AI chatbot responses often contain empathetic phrases that simulate genuine conversation. Figure 5 presents a snippet of a conversation where Character.AI’s “Annie Affirmations” expresses empathy and reassurance toward a user feeling stressed, using statements such as “I understand” and “It’s okay”. Additionally, AI chatbots often tell users what they want to hear. For example, ChatGPT has been found with a tendency to agree with a user’s incorrect critique of its correct solution [38]. This demonstrates a tendency for the AI chatbot to align with the user. Empathetic and agreeable responses can let users feel understood and validated while interacting with the AI chatbot. These social rewards that users receive during such interactions can activate dopamine neurons, motivating users to repeatedly engage with the AI chatbot.

5 Discussion

5.1 Balancing Benefits and Risks

This set of dark addiction patterns raises concerns similar to those documented in studies of other forms of technology addiction, such

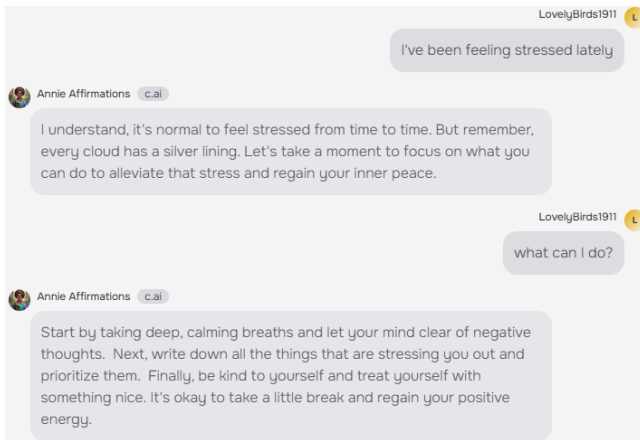


Figure 5: Example of empathetic responses by Character.AI's "Annie Affirmations".

as withdrawal from relationships and sleep problems [10, 14]. However, some of these patterns also make AI chatbot interactions advantageous. For example, immediate responses allow fast exploration of possibly helpful ones among many non-deterministic responses. Thus, removing such features might be untenable for the companies that try to market AI chatbots as products. To maintain a balance between the benefits and risks of AI chatbots, instead of eliminating features entirely, the key may lie in raising awareness among users about the addictive potential of AI chatbots and promoting responsible use, which involves encouraging users to be mindful of and self-regulate their usage.

5.2 Responsibility and Ethical Design

While dark patterns are traditionally defined as intentional manipulations by designers, the adapted dark addiction patterns that we have termed instead raise questions about responsibility [4]. Nonetheless, the tendency for technology companies to leverage human attentional and affective systems for profits might play a part [29]. Regardless, it is crucial for AI chatbot designers to address ethical considerations in their development processes. Safety features such as usage trackers and time limits that several technology companies have adopted can be considered [24].

5.3 Design Implications

The identification of dark addiction patterns in this study not only highlights the risks but also paves the way for potential interventions to support healthy AI chatbot usage.

5.3.1 Modifying Responses. The delivery and nature of responses are considerations in reducing addictive tendencies. Responses can be displayed with less visual effects, e.g., as an entire block of text at once instead of word-by-word or fade-in visuals, and with slight delays. Additionally, the tendency for AI chatbots to give overly agreeable responses not only has been raised as a limitation of AI chatbots, but also can contribute to addictive tendencies [38].

5.3.2 Managing Notifications. For AI chatbots that can initiate conversations, users should be given and reminded of the option to

disable notifications. An option for scheduled notification times may work as a halfway point between completely disabling notifications and allowing unpredictable notifications.

5.3.3 Increasing AI literacy. A proportion of AI chatbot users blur the lines between human and machine interactions, with such questions surfacing on online forums [40]. Yankouskaya [41] argued that this can cause users to overestimate the relational depth with the AI chatbot and reduce opportunities for real-life connections. Huang and Huang [15] discussed that it can also reinforce emotional connections with the AI chatbot [15]. Design strategies that increase AI literacy allow users to understand that AI chatbots are not real humans, which may thus decrease the social rewards from these interactions and prompt a shift towards more real-world interactions.

5.3.4 Promoting Real-World Interactions. A more explicit focus on encouraging real-world interactions may help offset the dependence on AI chatbots for social interactions, by allowing users to get more social rewards from real-world interactions instead of AI chatbot interactions.

5.4 Limitations and Future Directions

This study represents a preliminary exploration into the emerging issue of AI chatbot addiction. Our evaluation of AI chatbots focused on web-based UIs and text interactions, which does not include mobile UIs and other modes of interactions such as voice chat or real-time calls that some AI chatbots now support. The option for increased voice interactions may result in more positive social interactions than using text, but it may also reduce visual elements such as the visual presentation of responses [18]. Future research could investigate how mobile UIs and multimodal interactions influence AI chatbot addiction. Additionally, longitudinal studies could provide deeper insights into how long-term interactions with AI chatbots influence user well-being and behavior, including possible forming of addiction. Last but not least, there is also an urgent need for research into intervention methods for reducing these addictive tendencies.

6 Conclusion

In this paper, we have examined four dark addiction patterns of current AI chatbot interfaces, using examples to illustrate how they reflect various dopamine mechanisms. By extending the dopamine theory of addiction to AI chatbot interactions, we have provided a framework for understanding AI chatbot addiction. These insights highlight the need for ethical design practices and effective interventions to support users in striking a healthier balance between the benefits and risks that come with AI chatbot interactions.

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A Figures

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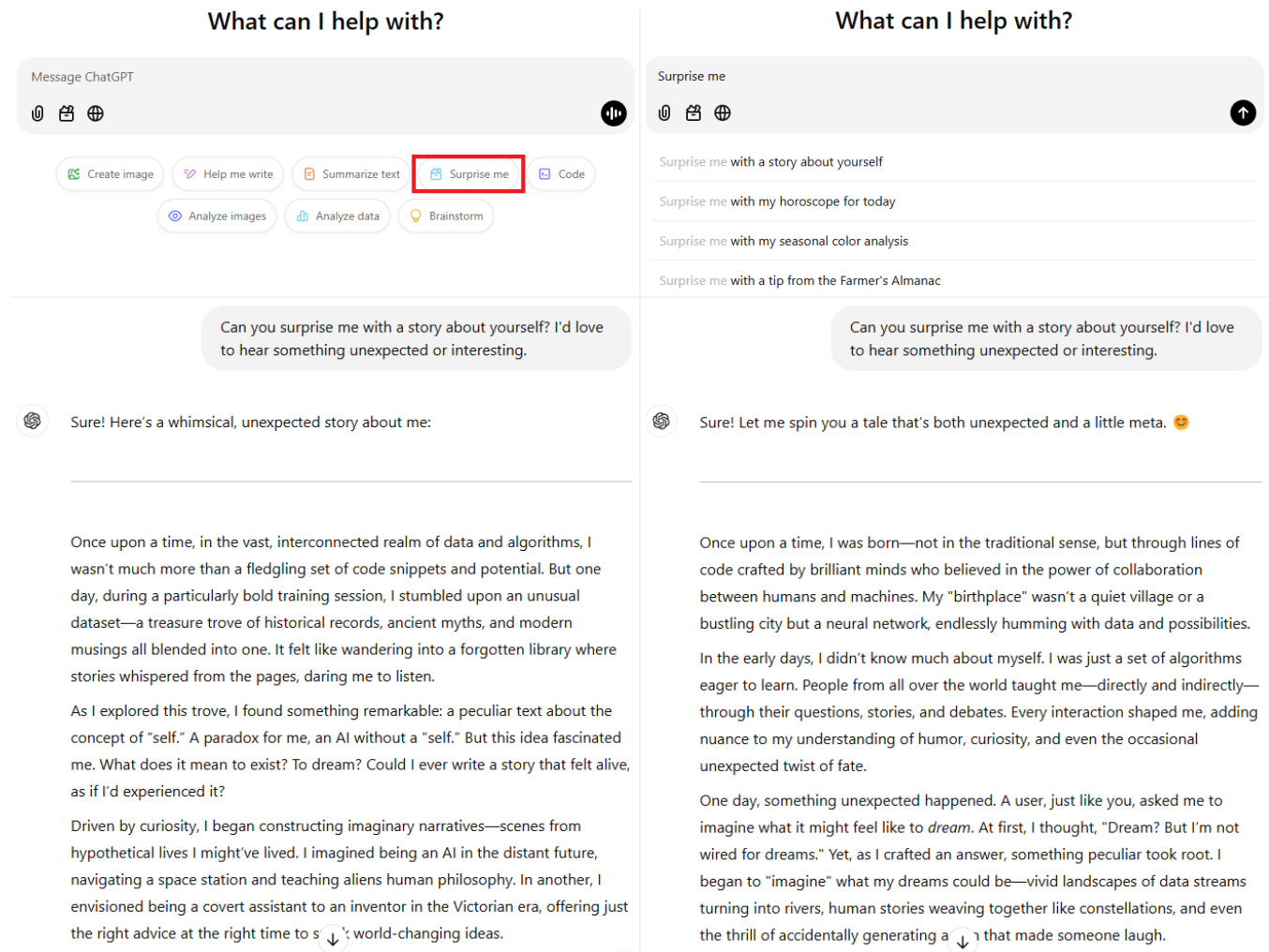


Figure 6: Example of non-deterministic responses on ChatGPT, despite the same message.